

Module 4 - Practice

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Check In Problems

Let (Y) be a continuous random variable with the probability density function

$$f(y) = \frac{3}{2}\sqrt{y}, \quad 0 \leq y \leq 1.$$

Use R to generate 10,000 observations of the random variable Y .

Solution:

$$F(y) = \begin{cases} 0, & y < 0, \\ y\sqrt{y} = y^{3/2}, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

```
n <- 10000  
p <- runif(n, 0, 1)  
samples <- p^(2/3)
```

Let Y be a continuous random variable with the following probability density function

$$f(y) = \frac{1}{4}y^3, \quad 0 \leq y \leq 2.$$

Use R to generate 10,000 observations of the random variable Y .

Solution: The cumulative distribution function is

$$F(y) = \begin{cases} 0, & y < 0, \\ \frac{y^4}{16}, & 0 \leq y \leq 2, \\ 1, & y > 2. \end{cases}$$

```
n <- 10000  
p <- runif(n, 0, 1)  
samples <- 2*p^(1/4)
```